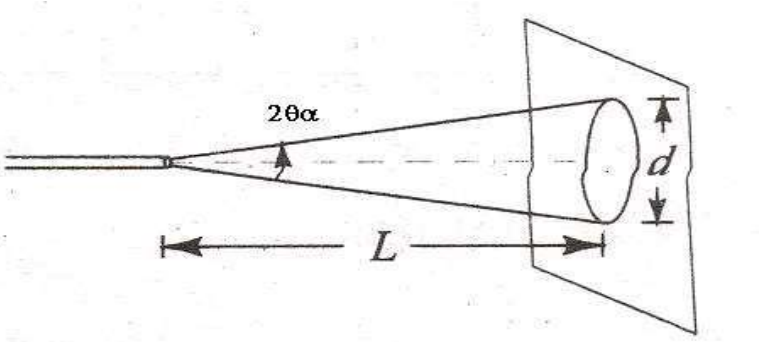
IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: To measure the numerical aperture of a step-index optical fiber.		
EXPERIMENT NO. 09	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 03

AIM: To measure the numerical aperture of a step index optical fiber.

APPARATUS: Optical bench optical fiber, diode laser, microscopic objective and a screen with a graph paper pasted on it

DIGRAM:



THEORY AND FORMULA USED:

To determine the Numerical Aperture (NA), the acceptance angle θα of the fiber should be known. The sine of the acceptance angle is called the numerical aperture.

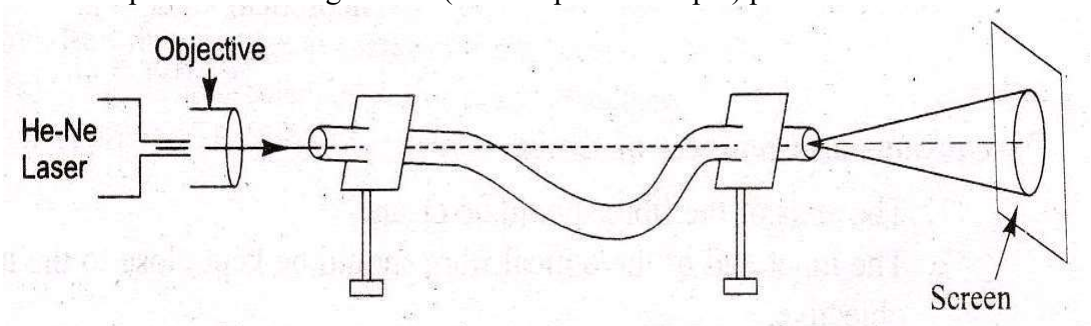
If the emerging rays from the output end of the optical fiber make, a spot of diameter d on a screen kept at a distance L from the output end of the optical fiber, then as it is obvious from Figure

Numerical Aperture (NA)

$$\mathbf{NA}=\frac{d}{\sqrt{d^2+4L^2}}$$

PROCEDURE:

1. Set up the apparatus as shown in Fig 2. Arrange the laser source, the microscopic objective, input and output end of the optical fiber at the same horizontal level with the help of stands.
2. Switch on the laser and make the light passing through the objective and the optical fiber fall on the screen.
3. Place the screen at some distance from the output end of the fiber such that it is perpendicular to the axis of the output end of the fiber. A circular spot is formed on the screen.
4. Mark the point at random (where the x and y axis crosses at a thick line on graph paper) and coincide the center of this circle with the marked point.
5. Gently move the optical fiber end away from the screen and confirm the symmetrical expansion of the circle's perimeter having a fixed (marked point in step 4) point of center.



6. Measure the distance L between the output end of the optical fiber and the screen. Also, mark and measure the diameter d of the spot on the screen.
7. Repeat the above procedure for different L values and measure the corresponding diameter of the spot formed.
8. Calculate the numerical aperture (NA) for each value of L and take the mean.

OBSERVATIONS:

1. Wavelength laser λ=
2. Position of the detector/screen on the bench (b) =…….

OBSERVATIONS TABLE:

Sr. No.	**Distance L(cm)**	**Distance d(cm)**	**NA** $= \frac{d}{\sqrt{d^2+4L^2}}$	**Mean NA**
1. **2.** **3.** **4**				

CALCULATIONS:

The NA formula is given below:-

$$\mathbf{NA} = \frac{d}{\sqrt{d^2+4L^2}}$$

RESULT:

The numerical aperture for a given optical fiber =……

PRECAUTIONS & SOURCES OF ERROR:

1. The ends of the fiber should be clean.
2. The input end of the optical fiber should be kept close to the microscopic objective.
3. The laser source should be kept close to the microscopic objective.
4. The plane of the output end of the optical fiber should be normal to the laser source.
5. The screen should be held vertically.

VIVA QUESTIONS:

1. What is the acceptance angle of the optical fiber?
2. What is the working principle of the optical fiber?
3. What are the different types of optical fibers based on modes?
4. What are the roles of the different components in the setup?
5. What do you mean by step-index optical fiber?
6. Why laser is diverging during exiting from the optical fiber end?